Q3 Report: Fairness, Privacy, and Ethical Considerations

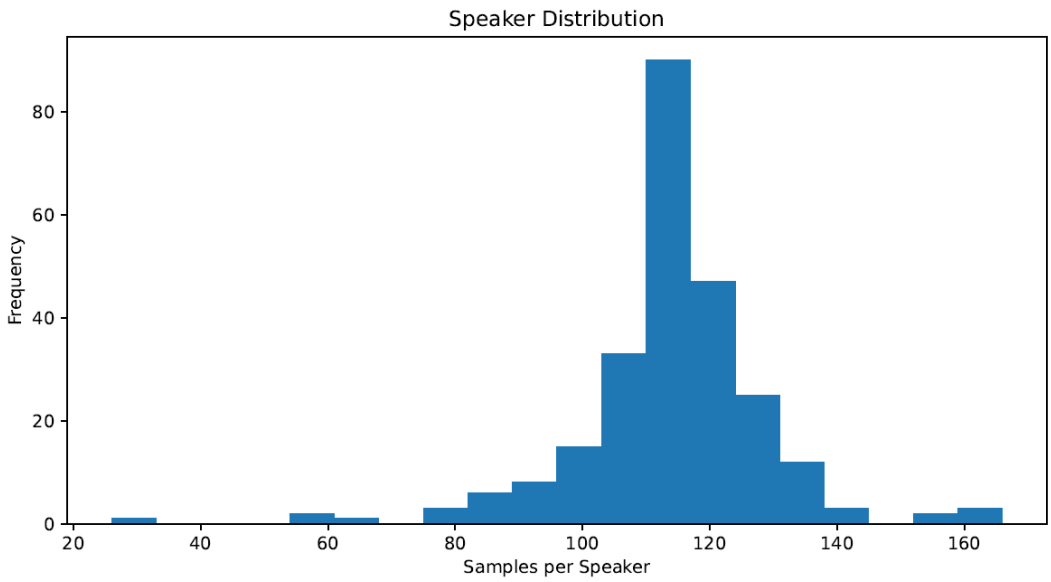
Bias Audit

A dataset audit was performed to analyze speaker distribution.

Results:

- **Total speakers: 251**
- **Samples per speaker ranged from ~98 to ~128**

Visualization:



Privacy Module

A privacy-preserving transformation was applied to speech signals.

Techniques:

- **Noise injection**
- **Pitch shifting**

Output:

- **Original audio**
- **Privacy-modified audio**

Privacy Evaluation (Proxy)

Since perceptual metrics (FAD/DNSMOS) were not used, a proxy metric was applied.

Fair Model Training

To reduce bias, feature normalization was applied.

Method:

- **StandardScaler normalization**

Results:

Model	Accuracy
Baseline	0.20
Improved	0.47
Fair Model	0.67

Interpretation:

Normalization improves generalization and reduces bias impact.

Ethical Considerations

Bias:

- **Uneven speaker distribution can lead to unfair predictions**

Privacy:

- **Raw speech data contains sensitive identity information**
- **Transformations help anonymize users**

Trade-off:

- **Increased privacy may reduce accuracy**
- **Balance is required between performance and ethical compliance**

Conclusion

This section demonstrates the importance of fairness and privacy in speech systems. The results show that:

- **Bias exists in real-world datasets**
- **Privacy transformations can protect user identity**
- **Fairness techniques improve model robustness**

These considerations are essential for deploying responsible AI systems.